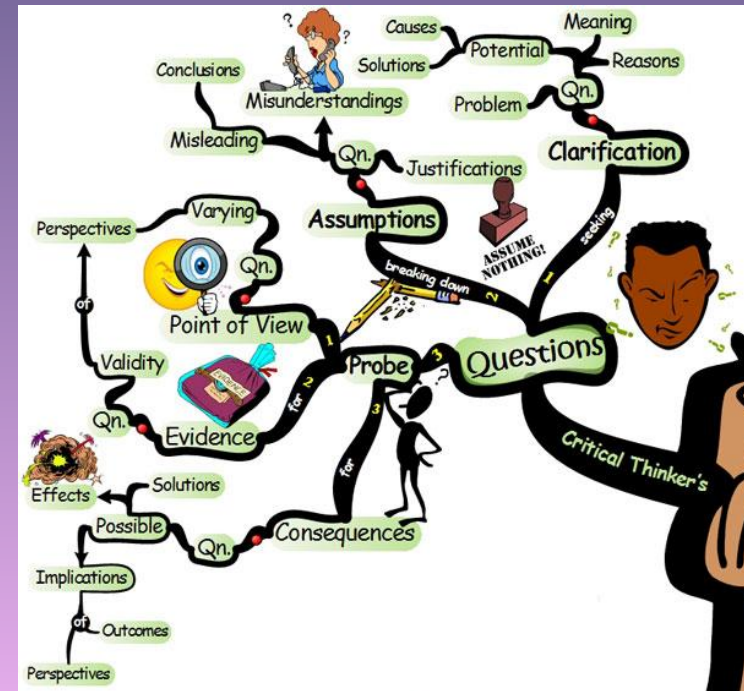


A Critical Thinker's Mindset

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Can you develop a critical thinker's mindset?

“The important thing is not to stop questioning. Curiosity has its own reason for existing.”
- Albert Einstein



Critical Thinking

Critical thinking is a kind of thinking in which you question, analyze, interpret, evaluate and make a judgement about what you read, hear, say, or write. The term critical comes from the Greek word kritikos meaning “able to judge or discern”.

Critical Thinking is NOT
driven by **answers**;
It IS driven by the
questions that you ask.

A Critical Thinker's Mindset

A way of thinking that involves an open, curious, and systematic approach to understanding, analyzing, and evaluating information and situations. It is a disciplined way of processing information and making decisions that is grounded in logic, evidence, and reasoning.



Developing a Critical Thinking Mindset

- Assess trustworthy information
- Develop mental agility
- Become a quantum thinker
- Avoid overthinking
- Balance rational, strategic, and emotional thinking

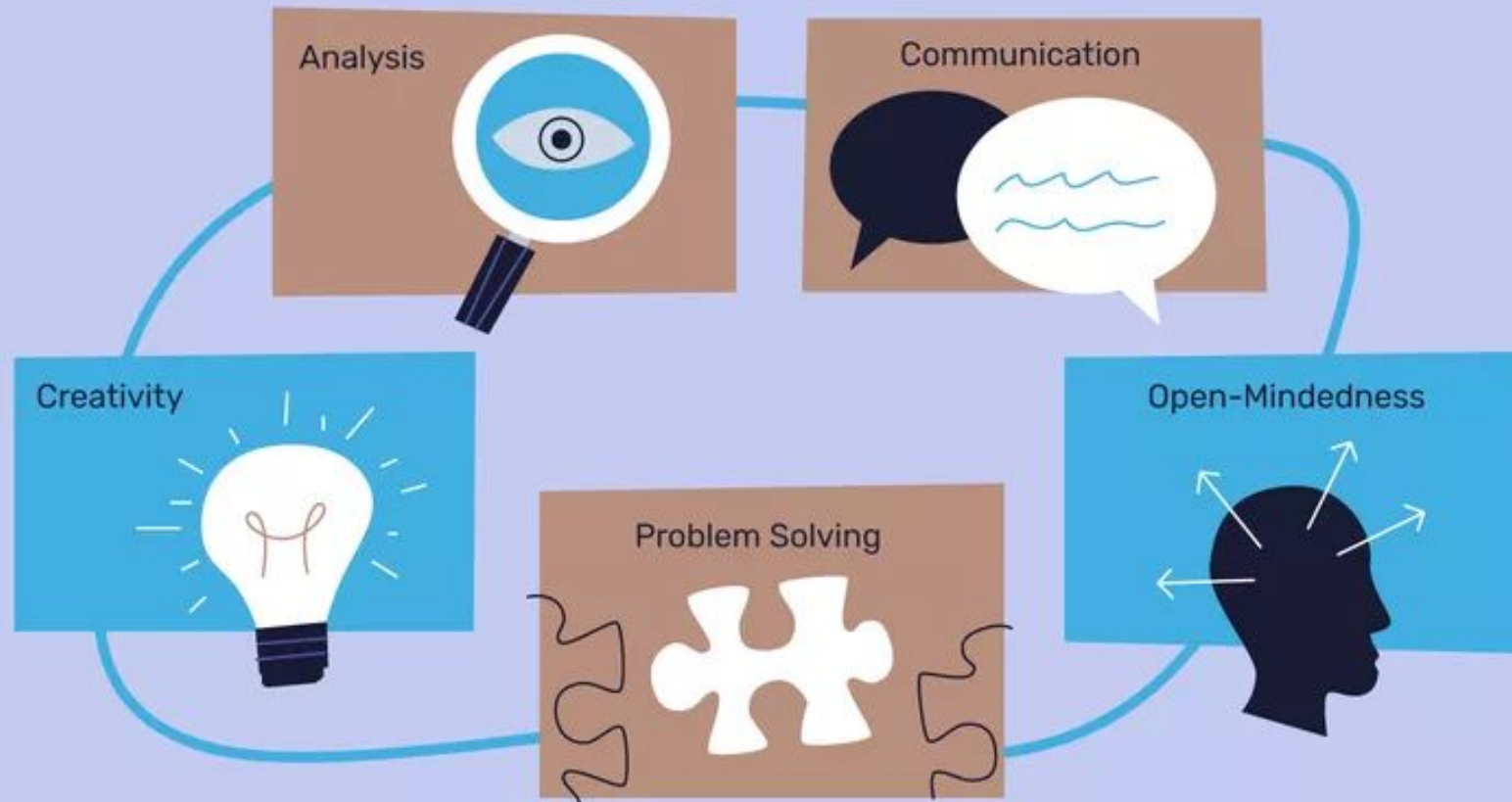
Importance of a Critical Thinker's Mindset

- **Decision-Making:** Informed decisions require careful analysis and reasoning.
- **Problem-Solving:** Identifying and solving problems effectively relies on evaluating different solutions.
- **Creativity:** Critical thinking encourages thinking outside the box and innovation.
- **Academic Success:** Academic work demands clear, logical arguments and the ability to critique sources.
- **Personal Development:** A mindset focused on critical thinking fosters personal growth and resilience.

Examples of a Critical Thinker's Mindset

- Science:** How critical thinking leads to breakthroughs in research.
- Business:** The role of critical thinking in strategic planning and risk management.
- Law:** How lawyers use critical thinking to build and defend cases.

Critical Thinking Skills



Criteria of Critical Thinker

- Left brain user.
- Using sequence and logic.
- Thinking before giving justification and result.
- Thinking carefully and take into consideration of risks and effects.

Characteristics of a Critical Thinker

- Curiosity:** A desire to learn more and explore beyond the obvious.
- Example:** Always asking "why" and "how" when confronted with new information.
- Skepticism:** Not accepting information at face value without questioning.
- Example:** Challenging sources and seeking corroborating evidence.
- Objectivity:** Analyzing situations without personal bias or emotional influence.
- Example:** Evaluating both sides of an argument equally before forming an opinion.

Characteristics of a Critical Thinker

- **Humility:** Recognizing the limits of your knowledge and being open to new evidence.
- **Example:** Being willing to change your mind when presented with new, convincing information.
- **Open-Mindedness:** Considering alternative viewpoints and being receptive to different ideas.
- **Example:** Engaging in discussions with people who have different perspectives.

Constantly Ask Questions

Example: Suppose your city has recently seen an increase in the number of coffee shops opening.

- **Why?**

Why are so many coffee shops opening? Is it because more people are developing a habit of drinking coffee, or because students and professionals need work-friendly spaces?

- **What?**

What effect is this increase having on small tea stalls or existing cafes? Are they losing customers or adapting their menus?

- **Where?**

Where are these coffee shops opening? Only in the city center and upscale areas, or also in residential neighborhoods?

- **Who?**

Who is affected by this increase? Are local business owners, students, office workers, or landlords benefiting the most?

- **When?**

When did this trend start? Has it grown over the past five years, or just recently after people's lifestyle changes post-pandemic?

- **How?**

How can the coffee shop industry sustain this growth? By offering unique flavors, better prices, or creating

The Role of Cognitive Biases

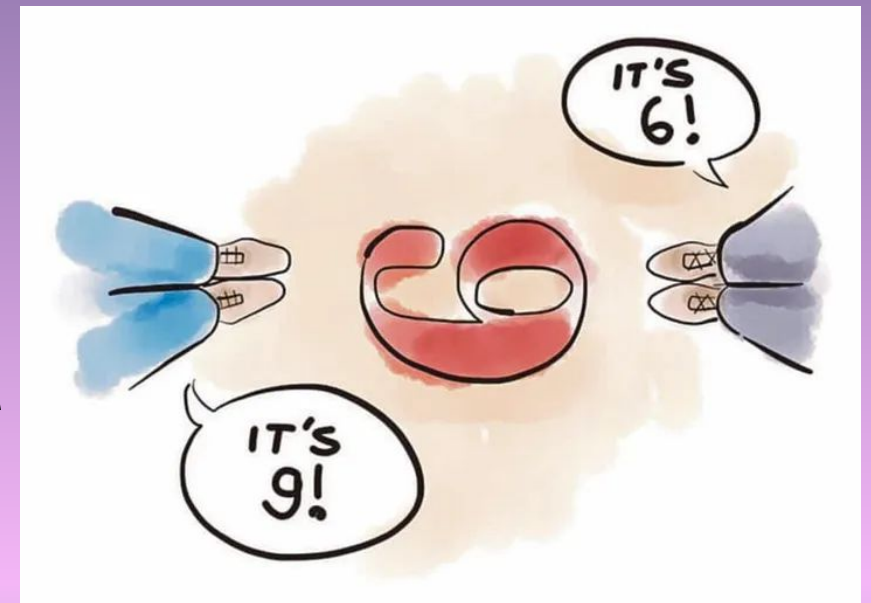
Definition: Cognitive biases are mental shortcuts that often lead to errors in thinking and judgment.

Examples:

- **Confirmation Bias:** The tendency to focus on information that confirms pre-existing beliefs.
- **Availability Heuristic:** Overestimating the importance of information that is most readily available.
- **Anchoring Bias:** Relying too heavily on the first piece of information encountered.
- **Overconfidence Effect:** The tendency to overestimate the accuracy of one's beliefs or predictions.

Confirmation Bias

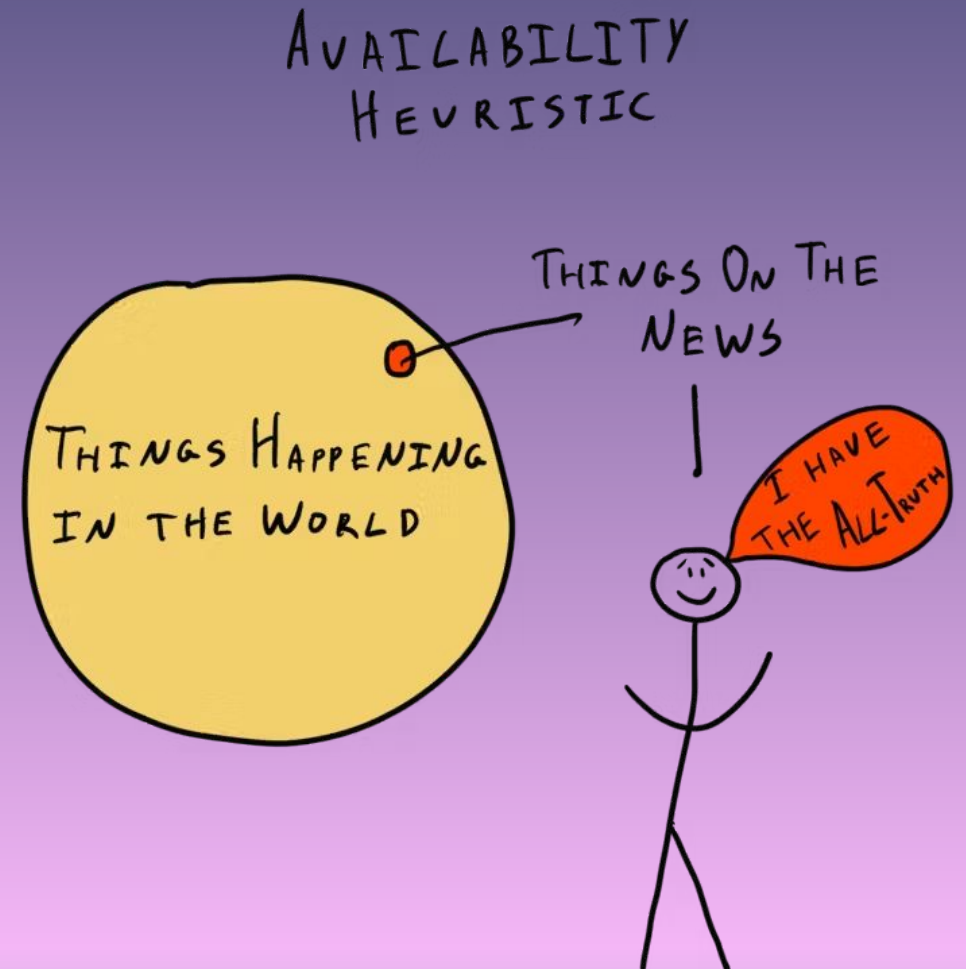
- **Example:** A person who believes that a specific diet is the best for weight loss selectively reads articles and watches videos that support this belief while ignoring studies that suggest other diets might be more effective or balanced. Even when presented with evidence to the contrary, they dismiss it, reinforcing their original belief without considering other possibilities.





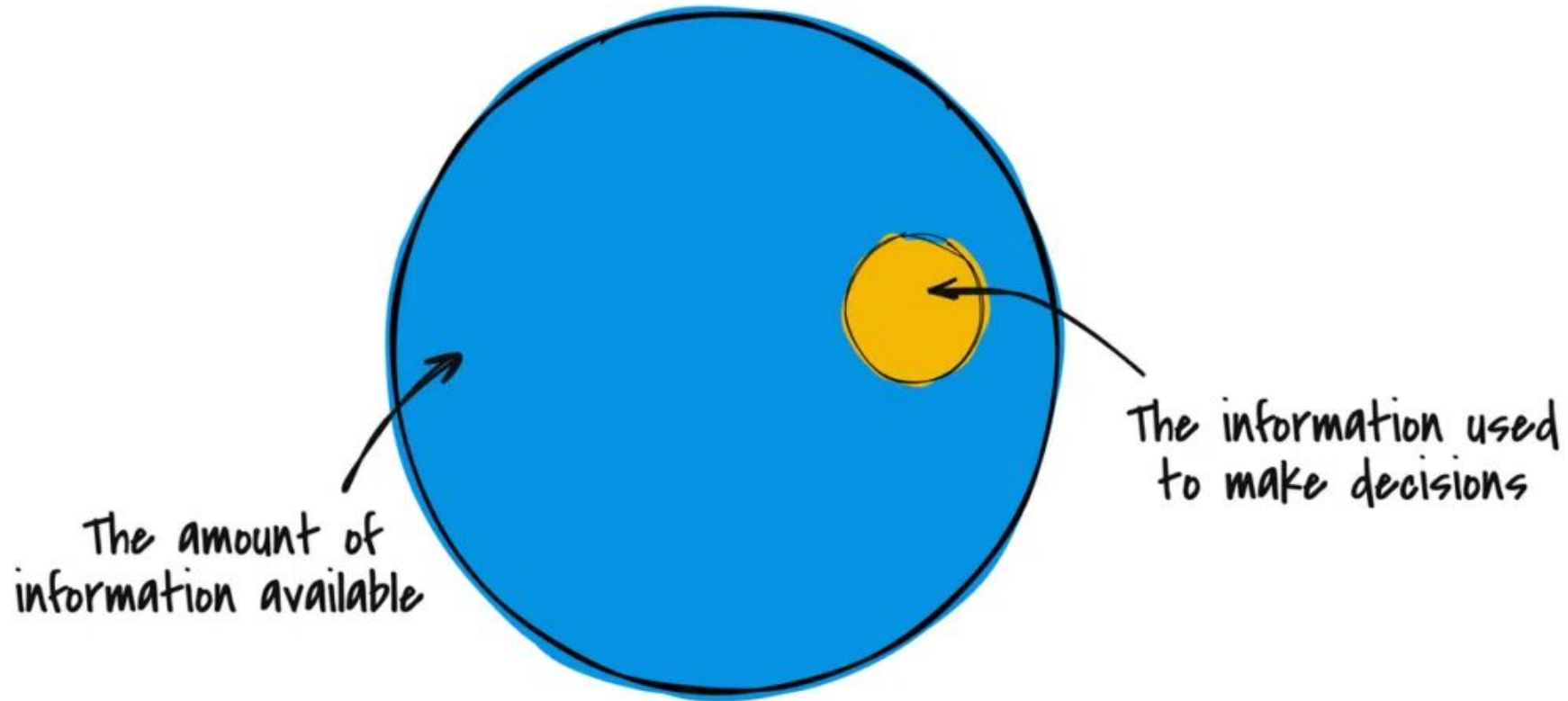
Availability Heuristic

- **Example:** After seeing news reports about airplane crashes, a person becomes fearful of flying, believing that air travel is highly dangerous. Despite statistics showing that flying is much safer than driving, the vivid and recent news coverage of plane crashes makes them overestimate the risk, leading them to choose less safe alternatives like driving long distances.



Availability Heuristic

Over reliance on 'top of mind' information because it's recent, frequent, or stands out.



There you go!!

Samsung Note 6



Samsung Note 7 banned
on all the flights

Radio

Another Note 7
catches fire

Anchoring Bias

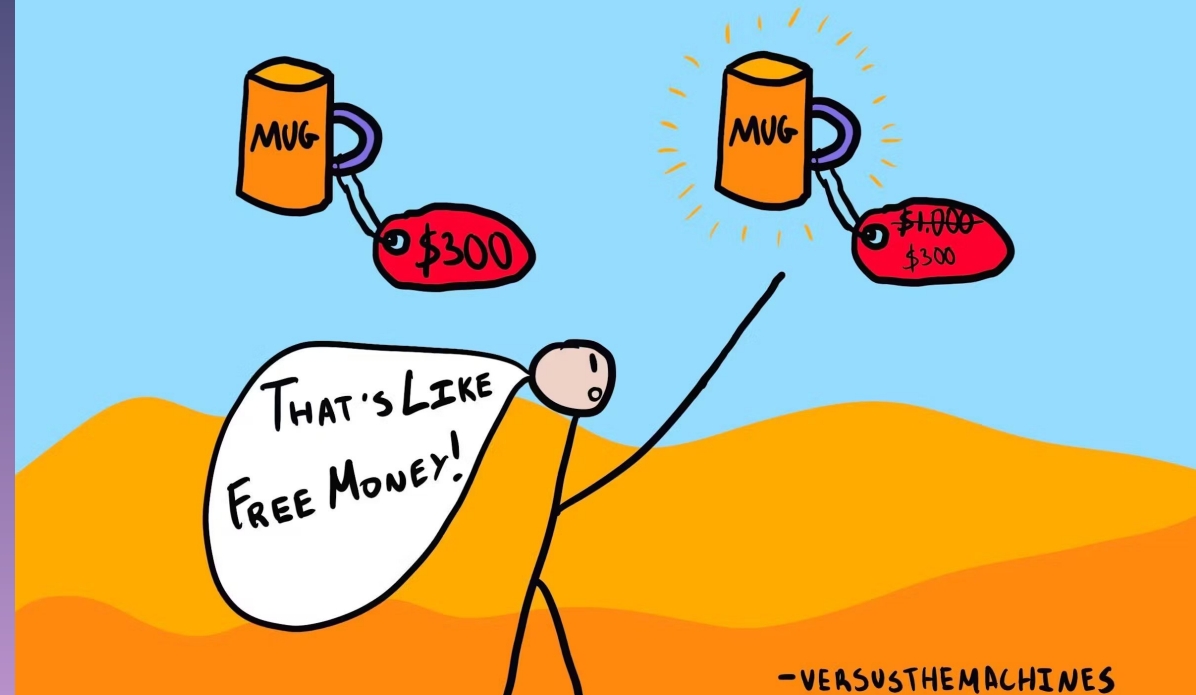
- **Example:** A customer shopping for a car is told the initial price of a vehicle is \$30,000. During negotiations, even when the dealer offers a discount, the customer remains anchored to the original price, perceiving the final price of \$28,000 as a good deal, even if it's still above market value. The first figure they encountered heavily influences their perception of what constitutes a fair price.



Anchoring bias binds you to your first impression



ANCHORING EFFECT



Overconfidence Effect

- **Example:** A stock trader is overly confident in their ability to predict market movements based on recent successes. They ignore warning signs and advice from colleagues, investing heavily in a risky stock. When the stock unexpectedly plummets, they realize they overestimated their knowledge and failed to account for the complete market.



Overconfidence Bias

The tendency of people to overestimate their abilities.



The Role of Cognitive Biases

Impact:

- **Decision-Making:** Biases can cloud judgment and lead to poor decisions.
- **Problem-Solving:** They can cause individuals to overlook better solutions.
- **Critical Thinking:** Recognizing and overcoming biases is crucial for developing a critical thinker's mindset.

Cognitive biases can hinder critical thinking and must be recognized and addressed.

Steps to Develop a Critical Thinker's Mindset

Step 1: Self-awareness:

Recognize Your Own Biases: Understand the biases you hold and how they influence your thinking.

Example: Consider a time when a bias influenced your decision-making.

Step 2: Active Learning:

Continuous Knowledge Seeking: Stay curious and open to learning.

Example: Regularly read articles, attend seminars, or engage in discussions that challenge your thinking.

Steps to Develop a Critical Thinker's Mindset

Step 3: Analytical Thinking:

Break Down Complex Information: Analyze information into smaller, understandable parts.

Example: When faced with a problem, identify its components and consider each separately before drawing a conclusion.

Step 4: Reflective Thinking:

Reflect on Your Decisions: Regularly review your thought processes and decisions to identify areas for improvement.

Example: After making a significant decision, reflect on whether it was the best course of action and what you could have done differently.

Steps to Develop a Critical Thinker's Mindset

Step 5: Practice: Engage in Exercises: Practice critical thinking regularly to strengthen your skills.

Example: Solve puzzles, engage in debates, or play strategy games that require logical thinking.

Practical Strategies

Strategy 1: Ask the Right Questions:

Who, What, Where, When, Why, and How?: Always ask these questions to gather comprehensive information.

Example: Before making a decision, ask yourself these questions to ensure you have all necessary information.

Strategy 2: Consider Multiple Perspectives:

Look at Issues from Different Angles: Understand how different perspectives can influence the outcome.

Example: In a debate, consider the opposing viewpoint and understand its logic before defending your own position.

Practical Strategies

Strategy 3: Avoid Assumptions:

Base Conclusions on Evidence: Avoid jumping to conclusions without sufficient evidence.

Example: When presented with a claim, ask for evidence before accepting it as truth.

Strategy 4: Practice Problem-Solving:

Engage in Regular Problem-Solving Exercises: Strengthen your ability to think critically by solving complex problems.

Example: Work through case studies or real-world scenarios that require critical analysis.

Practical Strategies

Strategy 5: Reflect on Outcomes: Learn from Past Decisions:

Regularly review your past decisions to identify areas for improvement.

Example: After a project is completed, reflect on what went well and what could have been done better.

Critical Thinking Skills Examples

- Problem-solving
- Creativity
- Decision-making
- Research
- Logical thinking
- Questioning
- Prioritization
- Leadership
- Strategy
- Planning
- Judgement
- Observation
- Active listening
- Analytical thinking

1) We think in order to decide what to do

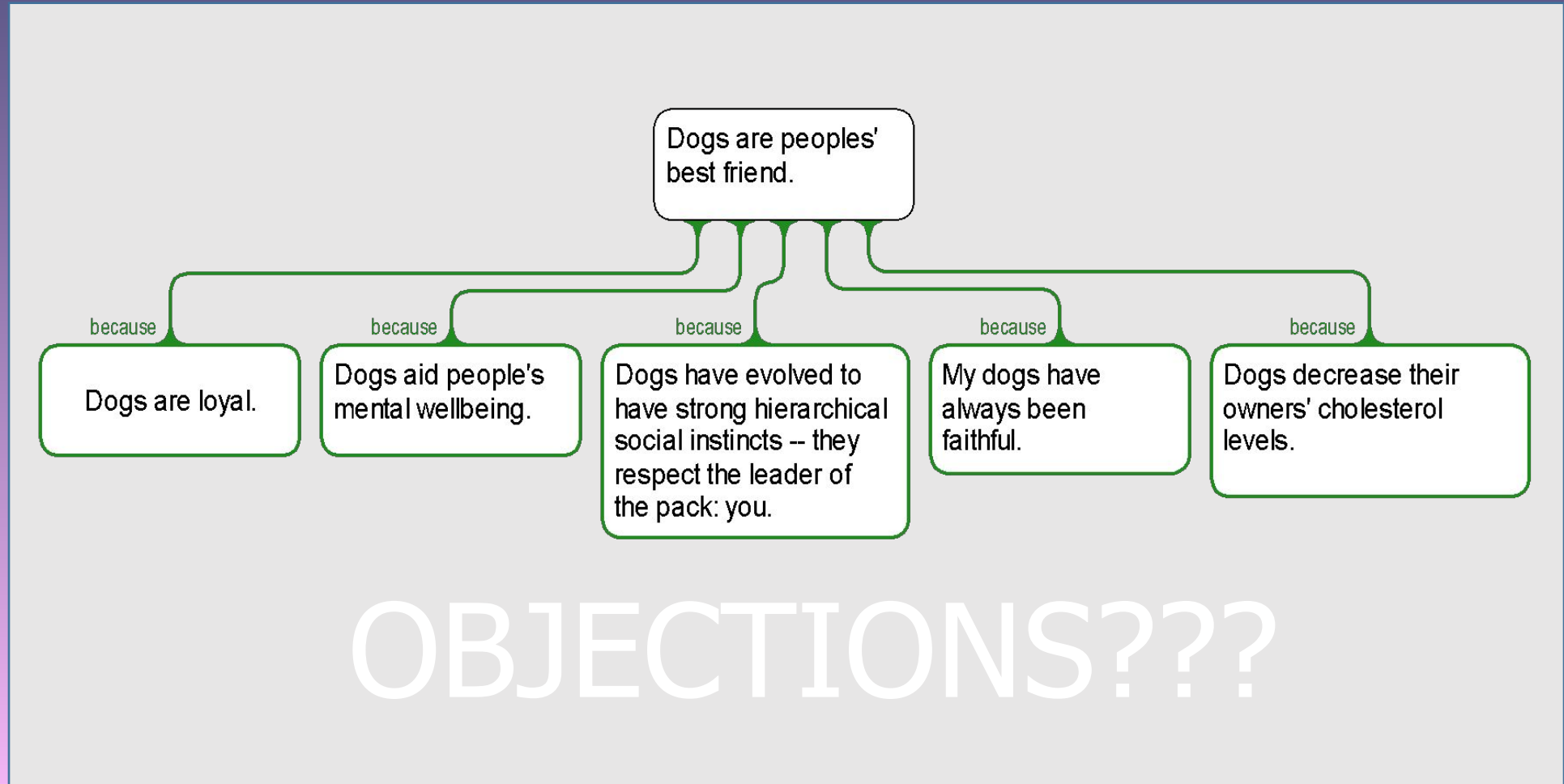
- Not only scientists think carefully and logically. We all do this whenever we care about our decisions.
- Consider an important decision:
“I should buy a dog”
 - Because I’ve always had dogs and I love them
 - Because dogs are peoples’ best friend
 - Because I can go out walking every evening, keep fit and meet other people with dogs
 - But walking my dog every evening will mean I cannot pursue my new hobby
 - But I’ll feel guilty if I’m forced to leave my dog alone in the house all day
 - But a new dog would be expensive and I’m really short of money right now.

How do we arrive at our final decision in this context?

2) We think in order to decide what to believe

- But our ultimate decision about what to do very often hinges upon our decision about *what we believe*. For example, what would make you believe the statement: Dogs are peoples' best friend?
- Consider a list of reasons

What to believe?

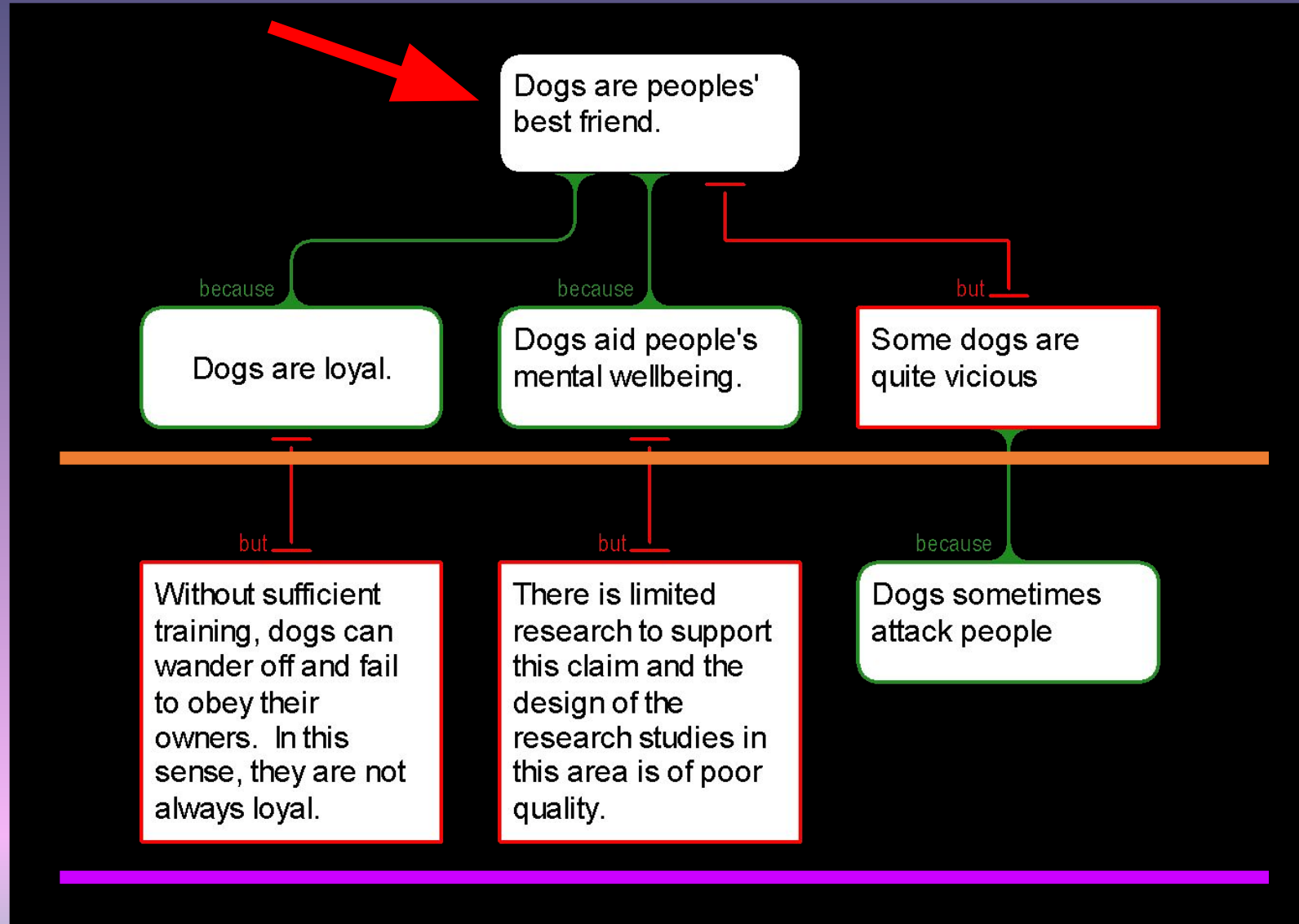


How do we ultimately decide what to believe in this context?

Questioning our beliefs: adding **but** to **because**.

- When deciding what to believe, we need to be careful not to focus only on reasons for accepting our beliefs.
- We should avoid simply working to confirm our beliefs (confirmation bias).
- We need to question our beliefs and the reasons we provide as a basis for our beliefs.
- We need to be sceptical.
- Are dogs really peoples' best friend. What would make you disbelieve?

Questioning our beliefs: adding **but** to **because**.



Consider a different belief (with the same argument structure as previous slide). Try adding **but** to **because**.

